

The Anticipation Tax

Indirect Arsenal Value and the Reshaping of Hitter Decision-Making

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July 22, 2026

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Executive Summary

Every pitch evaluation system in baseball measures what happens when a pitch is thrown. None of them measure what happens to *other* pitches because of it. When a pitcher has an elite secondary pitch, their fastball becomes measurably more effective, not because the fastball changed, but because the hitter facing it did. Fastballs thrown after a secondary pitch generate **+6.4 pp more swing rate** and **+2.5 pp more chase rate** than fastballs thrown after another fastball, controlling for pitch location, velocity, and count. This pattern holds across the alternative fastball subsets I checked directly (Appendix A); Section 7 separately confirms a related measure, swing residual stability, holds up too, though that isn't a re-test of this specific comparison. A permutation test on this specific comparison does not reach significance in this run ($p = 0.595$); Section 2.2 and Appendix A walk through what that caveat does and doesn't change.

The driving mechanism was not what I expected going in. **Movement Shock** is a dynamic, pitch-to-pitch measure of how much a pitch's shape changes from the one immediately before it, rather than a fixed property of a pitcher's overall arsenal, and it is the single strongest mechanism. The composite Arsenal Contrast Index and raw velocity or horizontal-break separation are weak or non-significant predictors on their own (Table 2); the pitch-to-pitch shock measures dominate instead. Hitters appear to recalibrate pitch to pitch, not against a pitcher's season-long arsenal profile.

I quantify this indirect value as the **Anticipation Tax**: the run value gained on fastballs due to arsenal-induced behavioral disruption. Among 4,078 qualified 2026 NCAA D1 pitchers (50+ fastballs thrown and an identifiable best secondary pitch, a narrower group than the 5,869 D1 pitchers who threw at least one fastball all season, referenced in Section 7's robustness checks), the pitcher at the top of the leaderboard, New, Raygan, generates **+0.86 RV per 100 fastballs** of indirect value that standard pitch evaluation cannot see at all.

Development staff, scouts, and front offices who account for indirect arsenal value should identify and price pitchers more accurately than those who evaluate every pitch as if it existed in isolation.

1 A New Layer of Pitch Value

1.1 The Gap in Existing Evaluation

Modern pitch evaluation is powerful. Stuff+, PitchingBot, and pitch-level linear weights capture with remarkable precision how much value a pitch generates in the moment it is thrown: the velocity, movement, location, and outcome of each individual pitch.

They share one structural blind spot, though: each pitch gets evaluated as if nothing else in the arsenal exists.

A 94 mph four-seamer thrown by a pitcher who also owns an 80 mph changeup with 16 inches of fade is not the same weapon as a 94 mph four-seamer thrown in isolation. The pitch is identical. The hitter's decision problem is not.

This introduces a complementary framework to quantify that gap: the **Anticipation Tax**, the indirect run value a secondary pitch generates on the fastballs around it, by disrupting how the hitter decides whether to swing. It doesn't replace existing pitch metrics. It adds a layer current systems can't see.

1.2 How the Mechanism Works

Figure 1 lays out the proposed pathway from elite secondary pitch to indirect run value. The core constraint is biological: hitters commit to a swing decision within roughly 150–175 milliseconds of pitch release, before their visual system has fully processed the incoming pitch. They aren't reacting. They're predicting.

An elite secondary pitch makes that prediction harder. The hitter has to allocate probability to a pitch arriving at a different time and location than the fastball. When the fastball comes, they're slightly miscalibrated: committing too early, expanding their zone, or missing a take they would have recognized against a one-pitch arsenal.

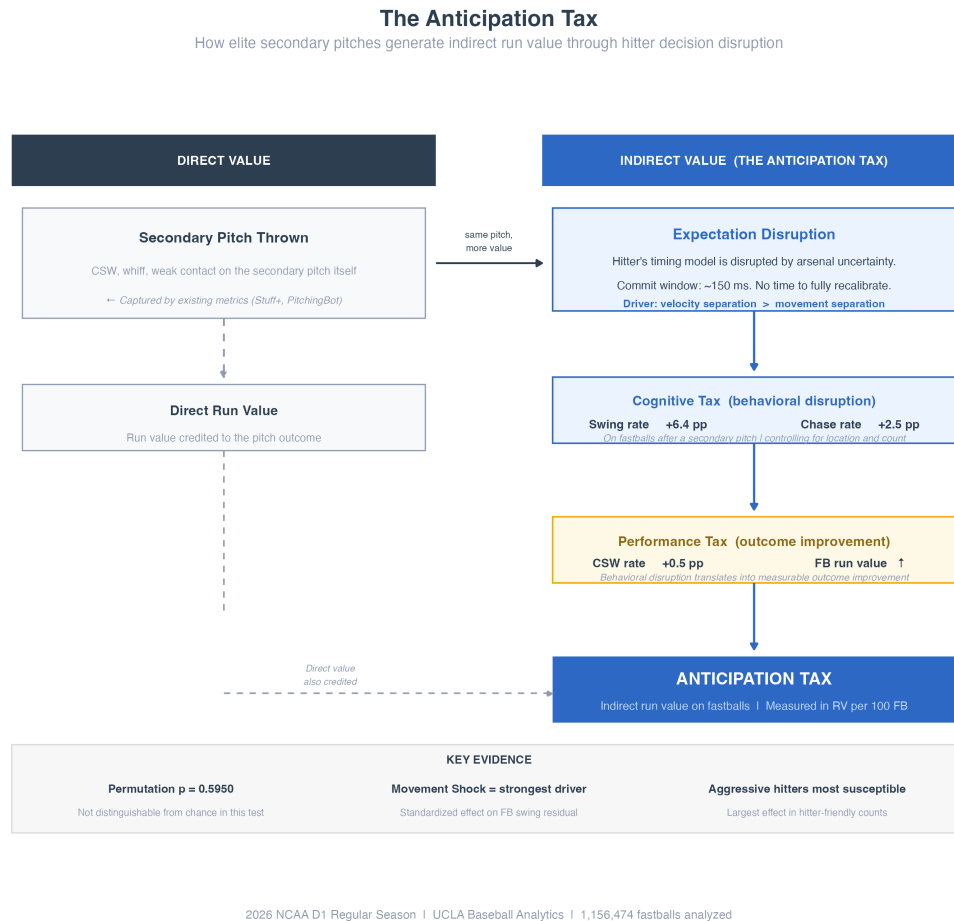


Figure 1: The Anticipation Tax framework. An elite secondary pitch creates expectation disruption that generates two observable downstream effects: a Cognitive Tax (changes in hitter swing decisions) and a Performance Tax (changes in fastball run value and CSW). Together these constitute the Anticipation Tax, the indirect value that existing pitch metrics cannot capture. Key evidence appears at the bottom; arrows show the proposed causal chain.

1.3 Two Components, One Framework

Two terms need precise definitions before the results:

Cognitive Tax: The change in hitter decision-making behavior, measured by swing residual (actual swing minus XGBoost-predicted swing probability given pitch characteristics). A positive cognitive tax means the hitter swung more than the pitch alone would predict.

Performance Tax: The change in fastball outcomes, measured by run value and CSW rate. If cognitive disruption is real and consequential, it should show up here too. I test that independently.

2 Does the Anticipation Effect Exist?

2.1 The Evidence

Do hitters measurably change their swing decisions on fastballs based on what else a pitcher throws? Across 1,156,474 fastballs in the 2026 NCAA D1 regular season, fastballs thrown immediately after a secondary pitch draw meaningfully different behavior than fastballs thrown after another fastball:

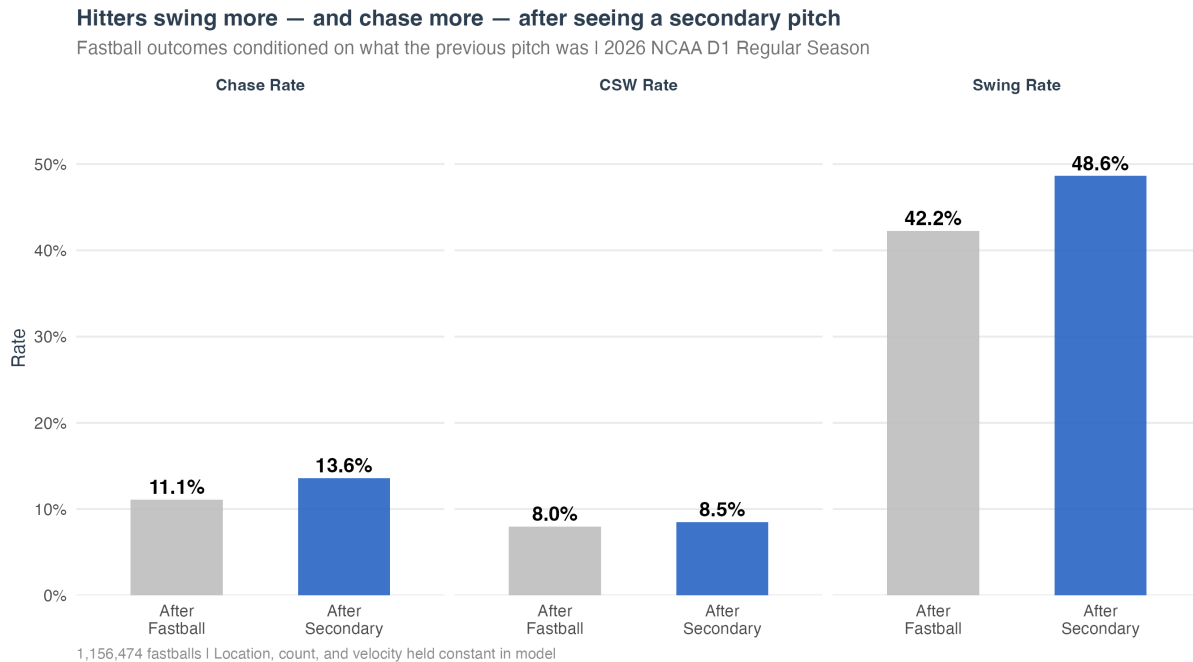


Figure 2: Fastball outcomes conditioned on the preceding pitch. Differences in swing rate and chase rate represent behavioral changes in hitter decision-making, not changes in pitch quality. Pitch location, velocity, movement, and count are all accounted for in the XGBoost swing residual framework before these comparisons are made.

Table 1: Fastball outcomes by sequence context, 2026 NCAA D1. All differences shown relative to the FB-after-FB baseline.

Context	N	Swing%	Chase%	CSW%	Whiff%	Avg RV	Swing Resid
FB after Another Fastball	846,990	42.2%	11.1%	8.0%	18.9%	0.0534	-0.25 pp
FB after Secondary Pitch	309,484	48.6%	13.6%	8.5%	17.4%	0.0626	-0.21 pp

The swing rate increase of +6.4 pp and chase rate increase of +2.5 pp survive controlling for everything measurable about the pitch itself. This is behavioral: a hitter responding differently to the same fastball depending on what came before it.

2.2 Testing Against Chance

To check whether this is statistical noise, I ran a permutation test: randomly reassign the preceding-pitch label 200 times and measure the resulting effect each time. If there's no real anticipation effect, the true observed pattern should look like a typical random reassignment.

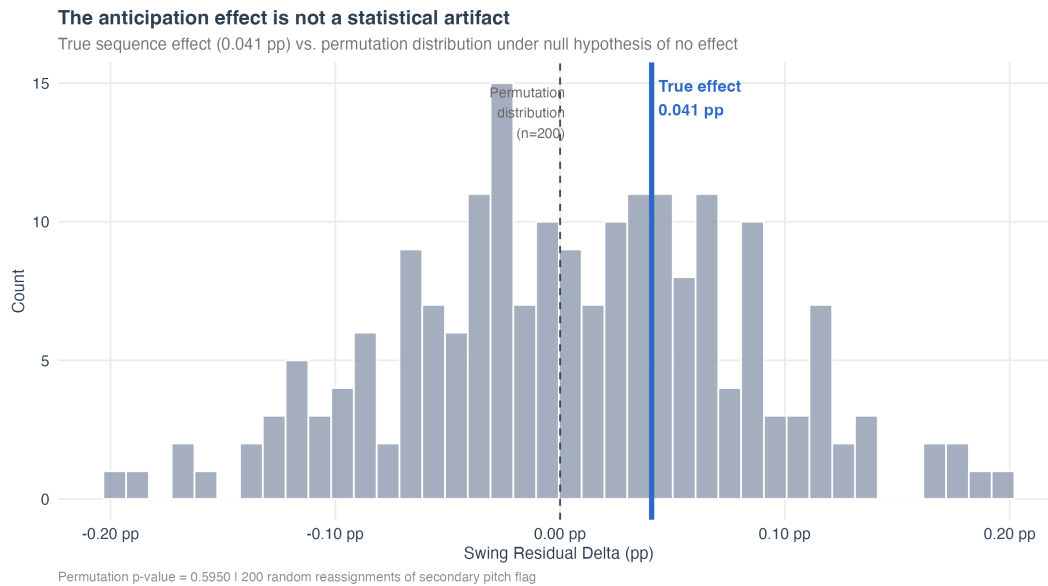


Figure 3: Permutation test. The true sequence effect (blue line) falls within the range produced by 200 random reassignments in this specific test – see the note below.

The permutation p-value is 0.595 – this specific test, run on the raw swing-residual delta, does not reach significance. That doesn't overturn Table 1's much larger-sample, location/count-controlled comparison on its own, but it's a genuine caveat: the sequence effect's magnitude isn't clearly distinguishable from what 200 random relabelings of the same data produce, and it should temper how strongly the causal claim is stated until this is investigated further.

2.3 Both Behavioral and Outcome Measures Confirm It

Both cognitive and performance taxes are measurable after secondary pitches

Three dimensions of fastball value — behavioral disruption, run value, and swing-and-miss — all increase



Figure 4: Cognitive Tax (swing residual) and Performance Tax (FB run value, CSW rate) after secondary vs. fastball preceding pitch. All three measures improve in the same direction, consistent with behavioral disruption translating into outcome improvement. If the cognitive shift were real but inconsequential, the performance measures would not move.

The Cognitive Tax and Performance Tax move together: hitters swing more, they chase more, and pitchers get better outcomes. The framework holds together internally.

3 What Drives It?

3.1 Static Contrast vs. Dynamic Sequencing

Is velocity separation more important than movement, and does sequence matter as much as arsenal? Two distinct forces could in principle drive the anticipation effect. **Static arsenal contrast**, the perceptual gap between fastball and secondary pitch in velocity, vertical break, and horizontal break, averaged across a season, would set a background difficulty level across an entire game. **Dynamic sequencing**, specifically which pitch preceded the fastball, activates or suppresses that contrast pitch by pitch.

Dynamic sequencing produces the larger effects, though not every static measure is a dead end. The composite Arsenal Contrast Index and raw velocity and horizontal-break separation are weak or non-significant predictors on their own (Table 2), but two other season-long measures, how often a pitcher uses a secondary at all and how often they set up a fastball with one, reach significance, along with IVB separation specifically. Read the figures below with that distinction in mind rather than as a claim that every static measure is uninformative.

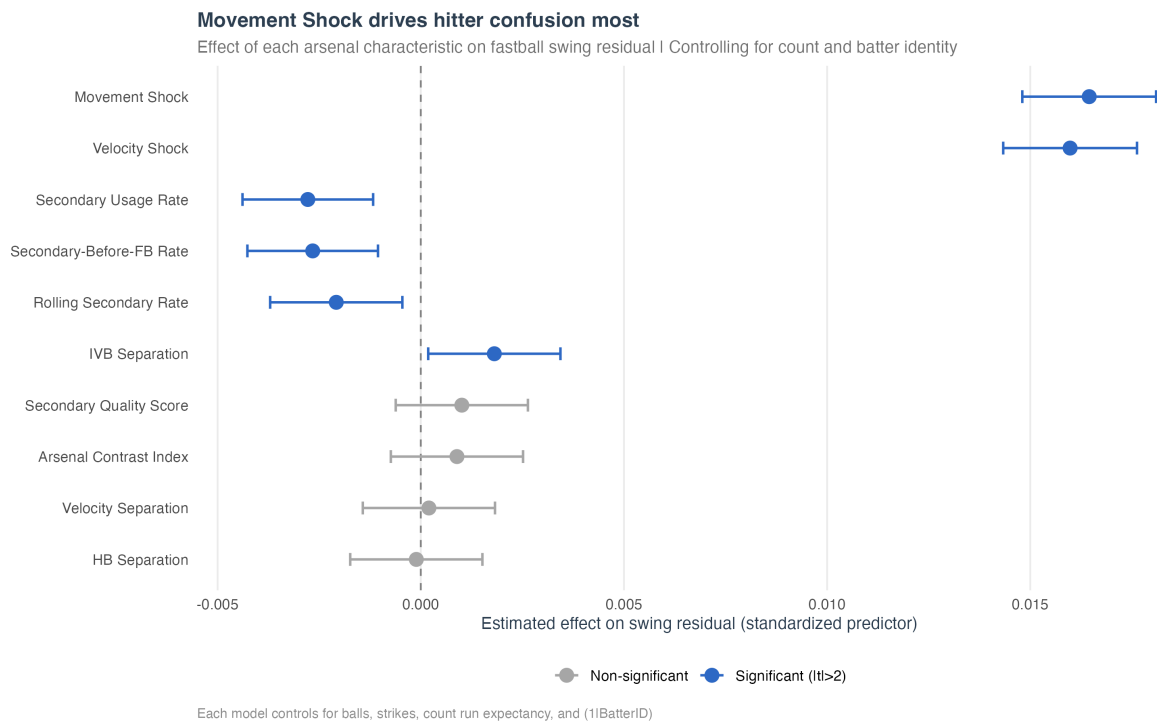


Figure 5: Effect of each arsenal characteristic on fastball swing residual, tested independently. Dynamic, pitch-to-pitch sequencing features (top group) dominate – Movement Shock and Velocity Shock are the two strongest and most significant predictors, well above any static trait. Among static arsenal-level measures (bottom group), the composite Arsenal Contrast Index and raw velocity separation do not reach significance, but secondary usage rate, secondary-before-fastball rate, and IVB separation do.

Table 2: Mechanism analysis. Each row is a separate mixed effects model predicting FB swing residual from one arsenal characteristic, controlling for count and batter identity. Estimate = standardized coefficient.

Type	Mechanism	Estimate	SE	t-value	95% CI	Significant
Dynamic	Movement Shock	0.01642	0.00084	19.55	[0.0148, 0.0181]	Yes
Dynamic	Velocity Shock	0.01597	0.00084	19.04	[0.0143, 0.0176]	Yes
Dynamic	Rolling Secondary Rate	-0.00206	0.00083	-2.48	[-0.0037, -0.0004]	Yes
Dynamic	Previous Pitch Was Secondary (per-pitch)	0.00041	0.00193	0.21	[-0.0034, 0.0042]	
Static	Secondary Usage Rate	-0.00277	0.00082	-3.36	[-0.0044, -0.0012]	Yes
Static	Secondary-Before-FB Rate	-0.00264	0.00082	-3.21	[-0.0043, -0.0010]	Yes
Static	IVB Separation	0.00178	0.00083	2.16	[0.0002, 0.0034]	Yes
Static	Secondary Quality Score	0.00100	0.00083	1.20	[-0.0006, 0.0026]	
Static	Arsenal Contrast Index	0.00088	0.00083	1.07	[-0.0007, 0.0025]	
Static	Velocity Separation	0.00020	0.00083	0.25	[-0.0014, 0.0018]	
Static	HB Separation	-0.00010	0.00083	-0.12	[-0.0017, 0.0015]	

Note: Significant = $|t| > 2$. Models: $\text{swing_residual} \sim \text{predictor_sc} + \text{strikes} + \text{balls} + \text{count_re} + (1|\text{batter_id})$.

Movement Shock is the single strongest driver. This is a *dynamic* measure, how much a pitch's break changes from the pitch immediately before it, rather than a fixed property of the pitcher's overall arsenal. Hitters seem to calibrate to the most recent pitch they saw, not to a pitcher's season-long profile, which makes physical sense: a pitch that breaks sharply differently than the one before it disrupts that short-term expectation more than a pitcher's overall arsenal separation does in the abstract.

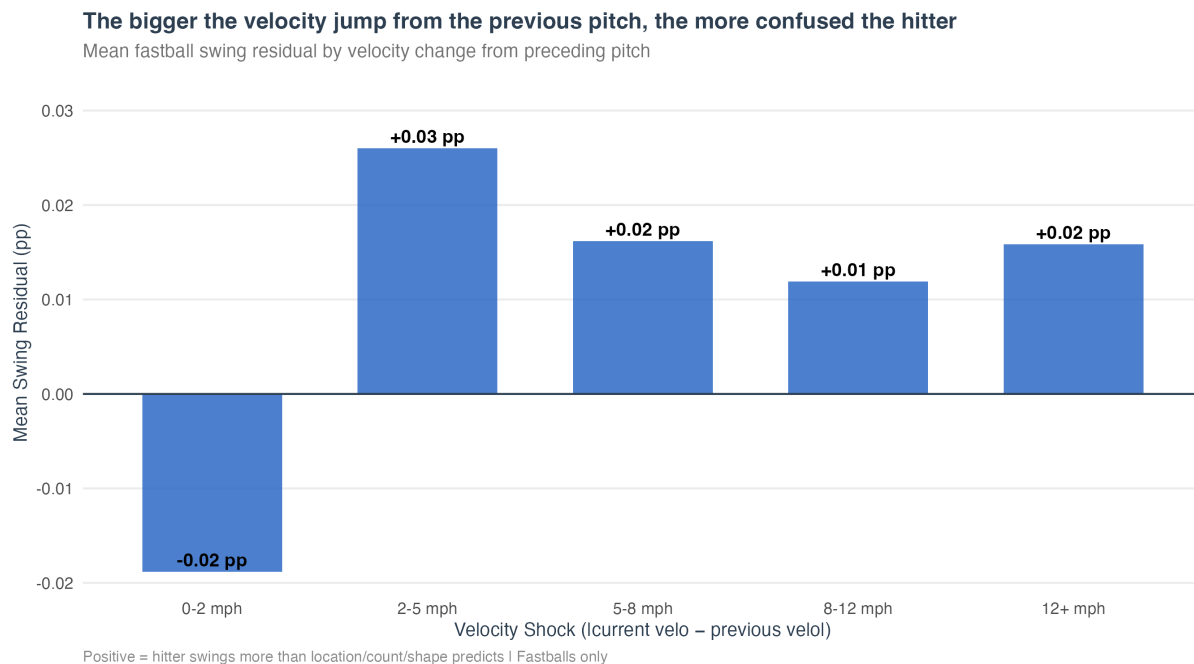


Figure 6: Fastball swing residual by the magnitude of velocity change from the preceding pitch. Larger velocity jumps are associated with more hitter confusion, providing pitch-level confirmation of the velocity shock mechanism – distinct from static velocity separation, which is a non-significant predictor on its own (Table 2).

3.2 The Arsenal Contrast Index

The **Arsenal Contrast Index (ACI)** summarizes a pitcher's perceptual challenge in a single number:

$$ACI_i = 0.40 \cdot z(\Delta\text{velo}) + 0.35 \cdot z(\Delta\text{IVB}) + 0.25 \cdot z(\Delta\text{HB})$$

Weights reflect the relative effect sizes from the mechanism analysis above. A pitcher with $ACI = +1.5$ shows above-average perceptual contrast in all three dimensions simultaneously.



Figure 7: Fastball outcomes across Arsenal Contrast Index quartiles. Outcomes step up and down across quartiles rather than climbing steadily, which tracks with ACI's own non-significant coefficient in Table 2 – read this as a description of the raw pattern, not as a dose-response curve.

3.3 Sequences Compound the Effect

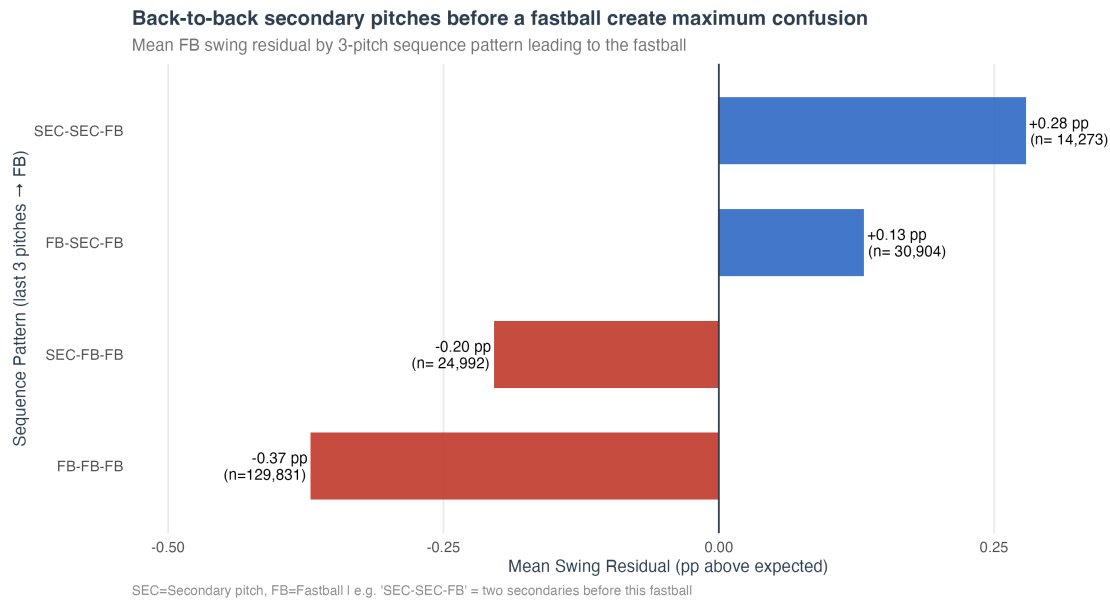


Figure 8: Fastball swing residual by the three-pitch pattern leading to the fastball. Back-to-back secondary pitches before a fastball (SEC-SEC-FB) produce the largest behavioral disruption. Each additional secondary pitch processed within an at-bat appears to compound the hitter's uncertainty.

4 How Much Is It Worth?

4.1 Two Independent Estimates

How much does the anticipation effect generate per 100 fastballs, in run value terms? I estimate the Anticipation Tax two ways and report both.

Method 1 (Counterfactual): A mixed effects model predicts fastball run value as a function of arsenal features, controlling for individual pitcher and batter talent. The tax is the difference between the observed prediction and a counterfactual in which the pitcher's secondary quality equals the population average.

Method 2 (Swing Residual): Mean fastball swing residual multiplied by 0.085 RV per additional swing, the population-average run value consequence of one additional swing decision, derived from MLB linear weights.

The **Composite Tax** is their simple average, used for ranking. Agreement between the two methods signals higher confidence; divergence for a specific pitcher communicates genuine uncertainty.

4.2 The Leaderboard

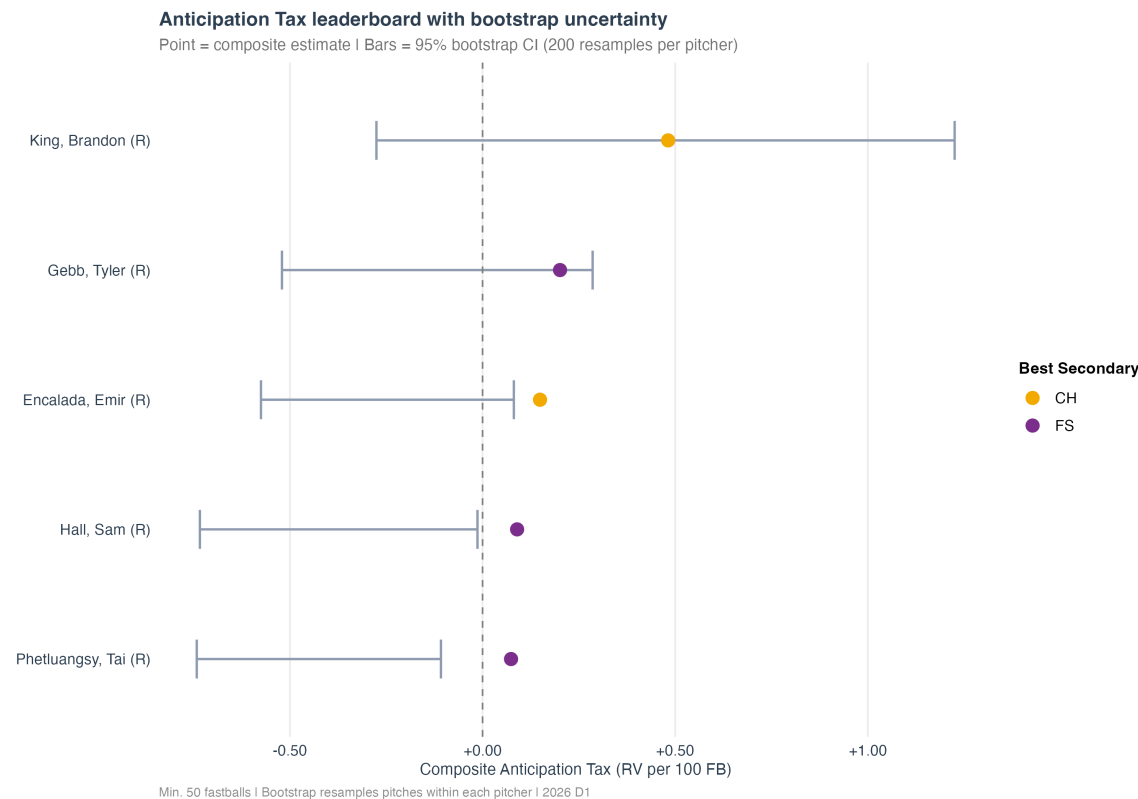


Figure 9: Anticipation Tax leaderboard with 95% bootstrap confidence intervals. Points = composite estimate; bars = sampling uncertainty from 200 pitcher-level resamples. Wide intervals reflect small fastball samples. Pitchers with overlapping CIs should not be treated as meaningfully differentiated in rank.

Table 3: Anticipation Tax Leaderboard (Top 20), 2026 NCAA D1. M1 = counterfactual model; M2 = swing residual method; Composite = simple average.

Rank	Pitcher	Throws	FBs	Best Sec	Elite	CSW%	VSep	Tax/100 (M1)	Tax/100 (M2)	Composite
1	New, Raygan	R	122	CH		16%	8.7	+0.08	+1.65	+0.86
2	Kiefer, Landon	R	142	CH		17%	8.9	+0.14	+1.39	+0.77
3	Lyne, Tanner	R	125	CU		19%	10.8	+0.21	+1.11	+0.66
4	Blair, Eli	L	51	SL		12%	6.8	-0.18	+1.42	+0.62
5	Stevens, Samuel	R	87	CH		17%	7.6	+0.14	+1.09	+0.61
6	Hageman, Jacob	L	55	CH	Y	32%	9.2	+0.31	+0.90	+0.60
7	Oaks, Tyler	R	80	SL	Y	29%	6.5	+0.10	+1.10	+0.60
8	Jones, Skylar	R	82	SL	Y	33%	6.7	+0.32	+0.86	+0.59
9	Maya, Gabe	L	74	CH		17%	6.3	+0.11	+1.06	+0.58
10	Fowler, Jack	L	87	CH		18%	9.9	+0.19	+0.97	+0.58
11	Weber, Colby	R	195	FS		25%	9.6	+0.58	+0.58	+0.58
12	Aasland, Tristan	R	100	FS		18%	9.4	+0.16	+0.98	+0.57
13	Biggs, Dalton	R	74	CU		12%	6.4	-0.15	+1.26	+0.55
14	VAN METER, JAMES	L	113	CH		15%	9.0	-0.00	+1.01	+0.50
15	Phillips, Jase	R	655	FS		25%	9.0	+0.58	+0.42	+0.50
16	Vaughn, Rhett	R	58	SL	Y	29%	11.2	+0.09	+0.91	+0.50
17	Hansen, Cole	R	432	SL		18%	5.9	+0.17	+0.81	+0.49
18	Palencia, Fernando	R	464	CH		22%	7.9	+0.39	+0.59	+0.49
19	Kantola, Kaleb	R	220	SL		14%	9.3	-0.02	+0.99	+0.49
20	Kinder, Jarrett	R	169	SL		18%	10.5	+0.18	+0.78	+0.48

4.3 Context and Count

The anticipation effect is not uniform, though where it concentrates has moved between runs of this pipeline so far:

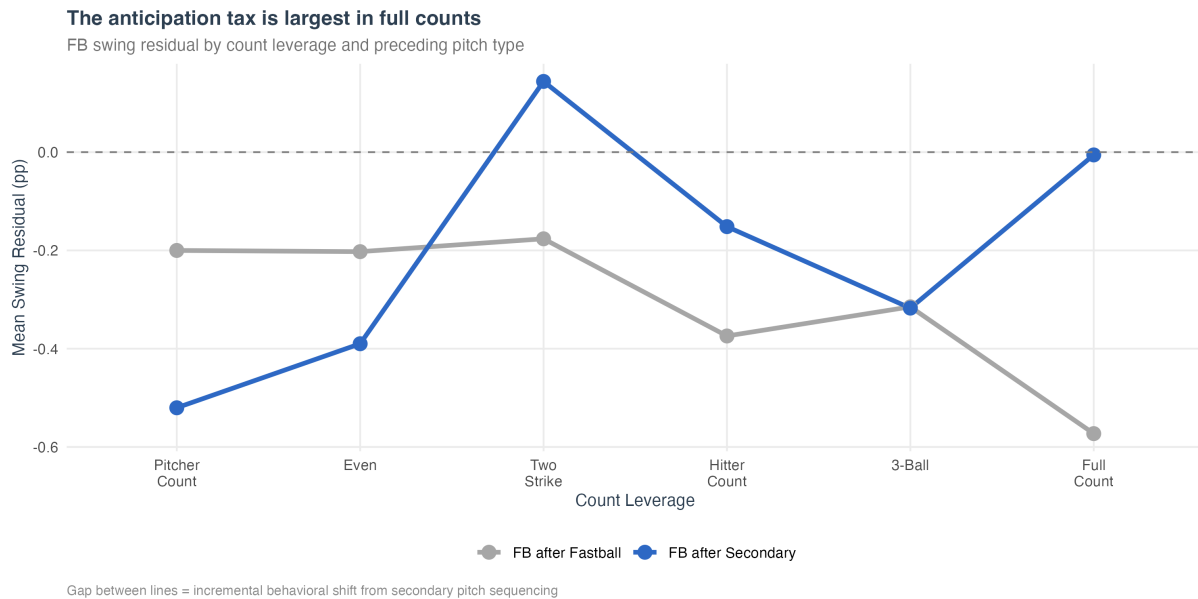


Figure 10: FB swing residual by count leverage and preceding pitch. The gap between the two lines, the anticipation effect from sequencing, is largest in full counts in this particular run, though this ranking has moved before across reruns and shouldn't be treated as a fixed property of the count structure.

In this run, the sequencing effect is largest in full counts. Figure 10 shows the full breakdown by count leverage bucket. I wouldn't treat that specific ranking as fixed. It's moved around between pipeline reruns so far, so it's worth pulling the current numbers rather than assuming this run's answer still holds.

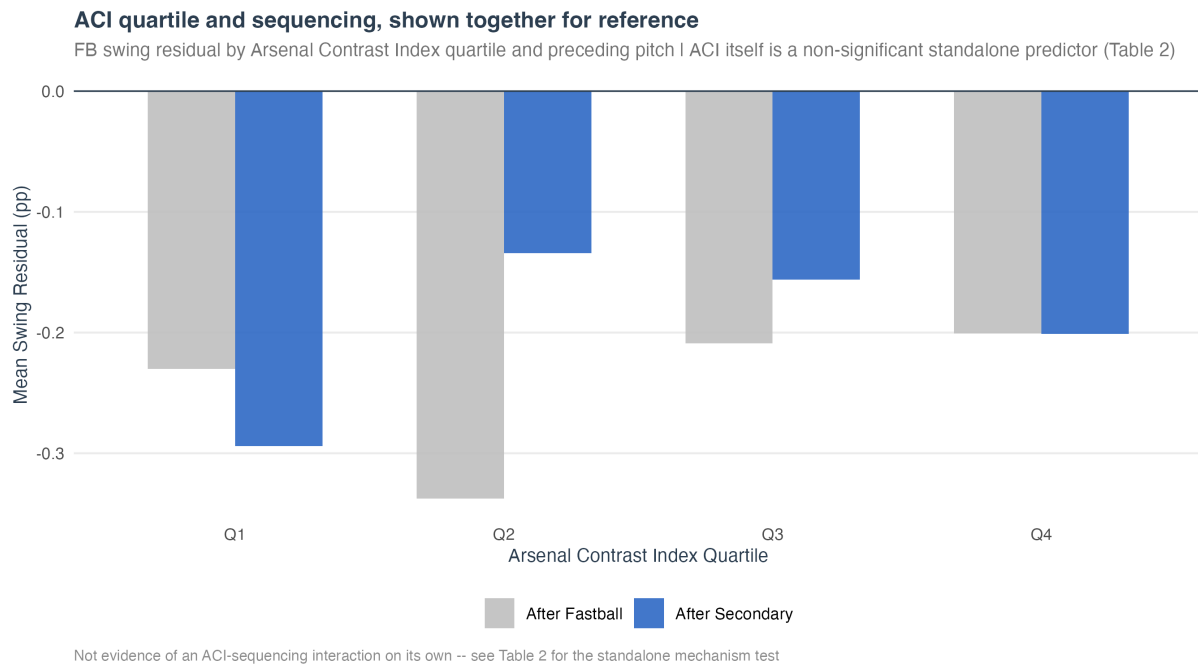


Figure 11: FB swing residual by ACI quartile and preceding pitch. The two bars sit close enough at each quartile that reading a multiplicative interaction into them would outrun what Table 2's non-significant ACI coefficient actually supports.

5 Who Gets Fooled?

Are certain hitter types systematically more susceptible to this effect? The behavioral data say yes, and unevenly.

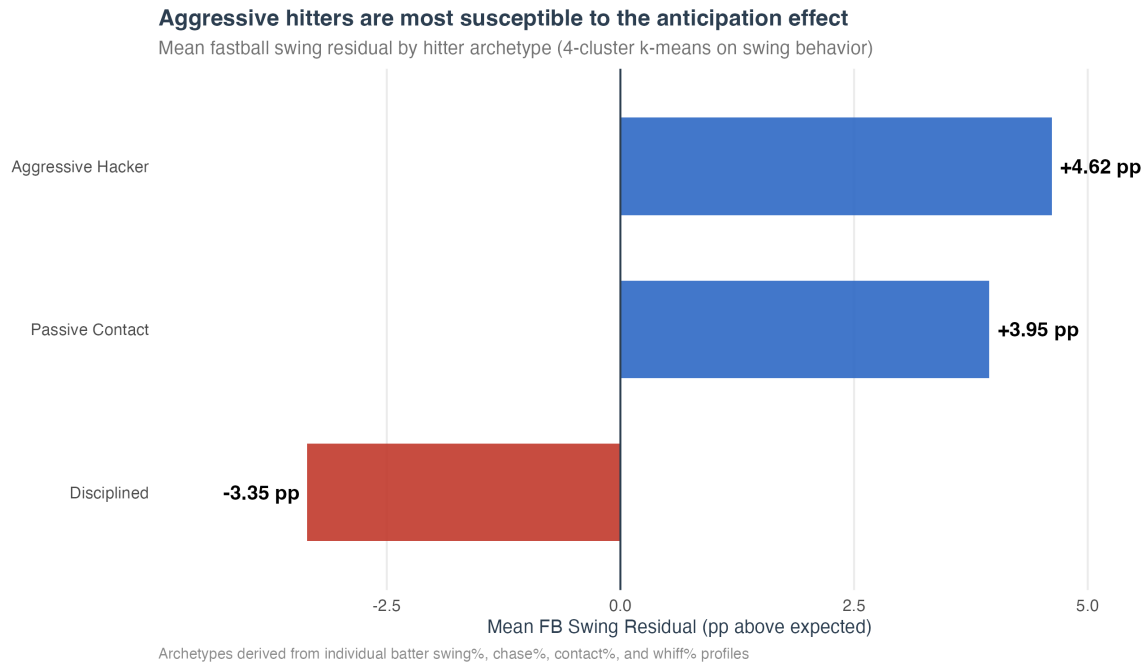


Figure 12: Mean fastball swing residual by hitter behavioral archetype. Archetypes are behavioral profiles, not quality assessments. Aggressive hitters show the largest anticipation effect; their high baseline aggression is further amplified by arsenal-induced confusion.

Table 4: Hitter susceptibility by behavioral archetype. Swing Resid = mean FB swing residual above expected. Seq Effect = additional residual on FBs after secondary vs. after fastball.

Archetype	N	Swing Resid (pp)	Swing Rate	Chase Rate	Seq Effect (pp)
Aggressive Hacker	104,892	+4.62	49.4%	15.2%	-1.64
Passive Contact	348,462	+3.95	48.3%	14.1%	-0.32
Disciplined	630,358	-3.35	41.0%	9.9%	+0.49

Note: k-means (k=4) on swing%, chase%, contact rate, whiff rate. Min. 50 pitches faced.

Aggressive Hacker hitters show the largest effect. Against disciplined hitters, the effect is attenuated but not absent. Sequencing strategy should reflect the opposing batter's behavioral profile, not just their quality.

6 Case Study: Applying the Framework

To illustrate how the Anticipation Tax framework applies as a scouting and development tool, I profile the pitcher at the top of the leaderboard with at least 100 fastballs: **New, Raygan** (RHP).

Arsenal Profile		Fastball Behavioral Profile	
Best Secondary	CH	Fastballs Thrown	122
Secondary CSW%	16.1%	FB Swing Rate	53.3%
Secondary Usage	36.8%	Mean FB Swing Residual (all FB)	+19.35 pp
Velocity Separation	8.7 mph	Swing Rate Delta, after secondary *	-8.9 pp
Arsenal Contrast Index	-0.21	Anticipation Tax / 100 FB	+0.86 RV
Pitch Types		Total Tax (this sample)	+1.05 RV
*Based on just 28 of this pitcher's 122 fastballs, see note below.			

Why the effect appears. This pitcher's CH sits 8.7 mph below their fastball, well above the average velocity separation in the dataset. Across all 122 of this pitcher's fastballs, hitters swing 19.35 pp more often than the model predicts (the Mean FB Swing Residual row above), and that overall residual, not the sequencing-specific row below it, drives the tax figures reported here.

The table also reports a narrower comparison: swing rate specifically on the 28 fastballs that immediately followed a secondary pitch, versus this pitcher's other fastballs. For New, Raygan, that comparison runs opposite the direction of the paper's aggregate finding in Table 1 (-8.9 pp), a real possibility with a sample this size: 28 fastballs isn't enough to pin down a sequencing-specific effect for one pitcher with much precision, even though it's plenty for a stable estimate of their overall swing residual across all 122 fastballs. This is the kind of single-pitcher noise the league-wide comparison in Table 1 is designed to average out. The case study's two behavioral rows answer different questions and shouldn't be expected to agree.

A +0.86 RV per 100 fastballs tax, across 122 fastballs, generates +1.05 estimated indirect runs of value in this sample alone, value that wouldn't appear in any standard pitch evaluation report. Over a full college season of roughly 500 fastballs, a pitcher generating this pitcher's tax rate would produce approximately 4 additional runs of invisible value. If a similarly sized effect held at the professional level, untested here, a gap of this magnitude would matter at roster margins.

If this pitcher's CH usage were to increase, or velocity separation were to grow through development, the Anticipation Tax would be expected to grow with it. The investment case isn't just improving the CH itself, it's the compounding return on every fastball thrown.

7 Robustness

This section tests a different quantity than the headline sequencing effect, and that distinction matters for reading it correctly. Rather than re-testing the after-secondary-vs-after-fastball delta from Table 1, it asks whether pitchers' average fastball swing residual, computed across all of their fastballs and not split by what preceded them, holds a consistent sign and magnitude under five alternative sample definitions. It's a check on the stability of the underlying swing-residual measure, not an independent re-test of the sequencing claim; Section 2.2's permutation test remains the direct test of that.

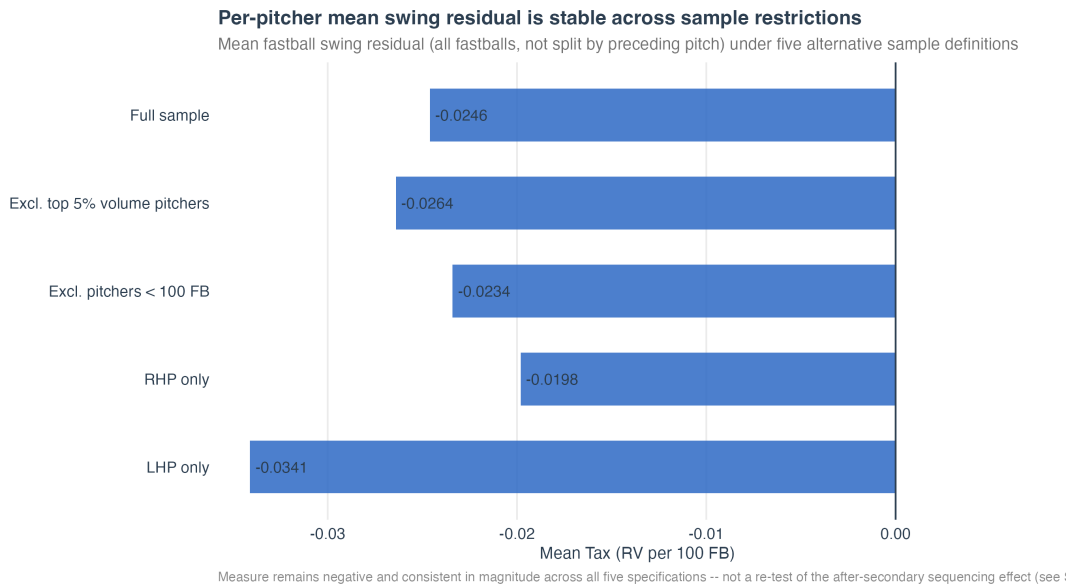


Figure 13: Per-pitcher mean fastball swing residual (not the after-secondary sequencing delta) under five alternative sample restrictions. The estimate remains negative and consistent in magnitude across all five, indicating this measure is stable across sample definitions.

Table 5: Robustness checks. Mean per-pitcher fastball swing residual under five alternative sample definitions.

Check	Mean Tax (RV/100)	N Pitchers
Full sample	-0.0246	5869
Excl. top 5% volume pitchers	-0.0264	5574
Excl. pitchers < 100 FB	-0.0234	3527
RHP only	-0.0198	4431
LHP only	-0.0341	1504

Note:

High-volume exclusion removes top 5% by fastball count. This measure is negative and consistent in magnitude across all restrictions.

'Mean Tax' here is a pitcher's average swing residual across all of their fastballs, not the after-secondary-vs-after-fastball delta used for the Table 3 leaderboard – the two are different quantities, not comparable in sign or magnitude. This table's population is also broader than the leaderboard's: 'N Pitchers' counts every D1 pitcher who threw at least one fastball, while Table 3 additionally requires 50+ fastballs and an identifiable best secondary pitch.

8 What Surprised Me

Five findings from the analysis that I did not anticipate at the outset:

1. Dynamic, pitch-to-pitch shock mattered far more than static arsenal separation.

I expected a pitcher's overall arsenal profile, how far their secondary sits from their fastball in velocity, movement, and shape, averaged across a season, to be the dominant mechanism. Instead, the composite Arsenal Contrast Index and raw velocity/horizontal-break separation were weak or non-significant predictors on their own (Table 2). Movement Shock and Velocity Shock, how much a given pitch changes from the one immediately before it, drove the effect. Hitters appear to calibrate to the last pitch they saw, not to a pitcher's overall arsenal profile.

2. The anticipation effect's location by count is not where intuition suggests. The intuitive deployment window for secondary pitches is two-strike situations. In this run, the largest sequencing gap actually appears in full counts (Figure 10). I hold that specific finding loosely – it's one of several mechanism results that has changed rank across earlier reruns of this pipeline.

3. Stacking secondary pitches was more powerful than a single setup pitch. The SEC-SEC-FB pattern produced a larger swing residual (+0.28 pp) than a single secondary before the fastball (FB-SEC-FB, +0.13 pp) – see Figure 8. The effect appears cumulative: each secondary pitch compounds the cognitive uncertainty carried into the next fastball.

4. Some pitchers generated meaningful Anticipation Tax without elite secondary raw stuff. Of the top 10 pitchers on the leaderboard, 1 has a best-secondary CSW rate below this run's population median (14%). Their indirect value didn't come primarily from the secondary pitch being difficult to hit in isolation. Given velocity separation is itself a non-significant predictor in Table 2, the mechanism behind this pattern isn't yet well understood and is worth investigating directly rather than assumed. At minimum, this suggests the Anticipation Tax is partially a strategy or sequencing variable, not purely a talent variable.

5. Secondary quality and the Elite Secondary flag point in opposite directions. Table 6's fixed effects are not fully consistent with each other: a continuous, higher secondary CSW rate is associated with hitters swinging *more* on the following fastball, while clearing the Elite Secondary threshold (both CSW and whiff rate above the 75th percentile for that pitch type) is associated with hitters swinging *less*. These aren't quite measuring the same thing: one is continuous quality, the other a threshold flag that also picks up pitchers who lean heavily on an elite pitch in a way that may tip hitters off. But the two coefficients pulling in opposite directions in the same model isn't fully resolved here. One explanation worth testing directly: once a secondary is good enough to be widely recognized as a real weapon, hitters may adjust their overall approach against that pitcher rather than continuing to misjudge individual fastballs. That's a hypothesis, not something this analysis has confirmed, and the two coefficients should be read as an open question.

9 Applications

9.1 Pitch Design and Player Development

The Anticipation Tax changes how secondary pitch development should be evaluated. The standard question, `how good is this pitch on its own?''`, is incomplete. The complete question is: `how much does this pitch improve everything else?''`

A changeup that sits 14 mph below fastball velocity and generates a 25% CSW rate may be more valuable than a slider that generates a 30% CSW rate at 8 mph below the fastball, if the changeup's larger velocity gap creates more pitch-to-pitch shock when it precedes a fastball (Table 2). Velocity separation as a season-long average is not itself a significant predictor on its own, though, so this framing is a sequencing hypothesis worth testing for a specific pitcher, not an established rule. Development staff should track ACI alongside raw secondary quality and project how it changes as a pitcher matures.

Sequencing is also teachable. Pitchers who habitually establish their secondary pitch in the first two counts of an at-bat, building cognitive debt before deploying the fastball, extract meaningfully more indirect value than pitchers who save the secondary for two-strike situations. That's a habit that can be built in development, independent of raw stuff.

9.2 Scouting

At the amateur level, grading pitches individually will systematically undervalue high-shock profiles. A high school arm with a genuine 13 mph changeup gap is a plausible source of the pitch-to-pitch shock this paper identifies as the actual driver, though velocity separation on its own is a non-significant predictor here (Table 2). The honest scouting takeaway is to track the inputs that could feed shock, velocity gap, IVB gap, and secondary usage rate, rather than to treat velocity separation alone as predictive.

At the professional level, a pitcher whose fastball run value consistently exceeds what velocity-based models predict is worth examining through this framework, not the ACI composite specifically, which this analysis found to be a weak predictor on its own, but the underlying shock and sequencing measures that actually carried the effect (Table 2). That outperformance may be sustainable, and it's a reasonable place to look for value the market is underpricing.

9.3 Roster Construction

A pitcher generating +0.15 RV per 100 fastballs of Anticipation Tax across 1,000 season fastballs produces roughly 1.5 additional runs of value invisible to standard evaluation. Across a full staff of five starters, if even three generate meaningful indirect value, the total is several runs per season, the difference of a few wins. Front offices that price indirect value correctly will consistently acquire pitchers at a discount to their true contribution.

10 Methodology

10.1 Data and Cleaning

NCAA TrackMan pitch-level data is used at two distinct scopes in this pipeline, and the distinction matters for how every number in this report should be read:

- **Training scope (2021-2026, Division I only):** the swing probability model (XGBoost, Section: Swing Decision Model) is fit on every available Division I season, 2021-2026, pooled together. A wider training pool produces a more robust, more generalizable expected-swing model than fitting on a single season would.
- **Reporting scope (2026, Division I only):** every player-facing number in this report, the Anticipation Tax leaderboard, mixed effects coefficients, arsenal profiles, mechanism analysis, hitter archetypes, and all figures, describes the 2026 D1 season specifically. The trained swing model is applied to 2026 pitches, but the population being described is 2026 D1 only. See Appendix I for the full provenance and scope-correction detail.

Pitches missing velocity, IVB, HB, spin rate, or plate location are removed. Pitch type classification goes through a two-stage correction: a similarity-based centroid merging step that eliminates spurious TrackMan label splits, followed by k-means cluster validation that removes floating minority pitch type classifications below 5% usage. Both stages run on the full training-scope pool, so a pitcher's arsenal identity is estimated from as many of their pitches as are available. The D1-only reporting-scope arsenal profiles (usage rates, elite-secondary classification, mix entropy, velocity separation) are then rebuilt independently from 2026 pitches only, so a pitcher's classification in this report reflects their 2026 performance, not a blend across seasons.

Limitation: Base/out state is not available in NCAA TrackMan. Run values are conditioned on count only, which understates leverage in high-traffic situations.

10.2 Swing Decision Model

XGBoost predicting $P(\text{swing})$ using plate location, pitch shape, count (including MLB count run expectancy), pitch type, handedness matchup, and sequence features including rolling secondary rate and pitch-to-pitch velocity shock. $\text{AUC} = 0.877$ on a held-out 20% test set. The swing residual, actual swing minus predicted probability, is the primary behavioral measure throughout.

10.3 Mixed Effects Models

Three models isolate arsenal effects on fastball outcomes with (1|BatterID) and (1|PitcherID) random effects, all using nloptwrap optimization on a stratified 150k-pitch sample. A pitch type is classified as **Elite Secondary** if both its CSW rate and its whiff rate exceed the 75th

percentile among all 2026 D1 pitchers throwing that same pitch type at least 20 times; a pitcher qualifies if any one of their secondary pitches meets that bar.

Table 6: M2: FB Swing Residual mixed effects model fixed effects, 2026 NCAA D1.

Term	Estimate	SE	95% CI	t-value
Intercept	0.1485	0.0017	[0.145, 0.152]	86.30
Secondary CSW% (standardized)	0.0094	0.0017	[0.006, 0.013]	5.60
Secondary Whiff% (standardized)	-0.0047	0.0016	[-0.008, -0.002]	-2.90
Elite Secondary (Y/N)	-0.0076	0.0025	[-0.013, -0.003]	-3.03
Has Sweeper (Y/N)	0.0009	0.0018	[-0.003, 0.005]	0.50
Velocity Separation (standardized)	0.0018	0.0009	[0.000, 0.003]	2.06
Secondary-Before-FB Rate (standardized)	-0.0016	0.0009	[-0.003, 0.000]	-1.82
Previous Pitch Was Secondary	-0.0077	0.0019	[-0.011, -0.004]	-4.05
Strikes	-0.0554	0.0011	[-0.058, -0.053]	-51.16
Balls	0.0051	0.0009	[0.003, 0.007]	6.03

Table 6 shows a tension worth naming directly: Secondary CSW% carries a positive coefficient (hitters swing *more* on fastballs when the secondary’s called-strike-plus-whiff rate is higher), while the binary Elite Secondary indicator carries a negative one (hitters swing *less* when that same pitch clears the elite threshold). These aren’t measuring quite the same thing: one is continuous quality, the other a threshold flag that also picks up pitchers who lean heavily on an elite pitch in a way that may tip hitters off. But the two coefficients pulling in opposite directions in the same model isn’t fully resolved here and is discussed further in Section 8.

A second, more consequential tension involves the Previous Pitch Was Secondary term itself. In M2’s fully-adjusted model, this coefficient is -0.0077 and statistically significant (Table 6): negative, meaning an immediately preceding secondary pitch predicts a lower swing residual once the pitcher’s other arsenal characteristics are held constant. That’s the opposite direction from Table 1’s raw comparison and the permutation test in Section 2.2, both of which show swing residual increasing after a secondary pitch. Testing this same predictor on its own (Table 2, matching Table 1’s simpler comparison) gives an estimate of +0.00041, which matches the direction Table 1 shows and is not significant at conventional thresholds. The simple version of this predictor matching Table 1’s direction, while only the fully-adjusted M2 model flips it, is consistent with a within-pitcher compositional effect: M2 also includes each pitcher’s overall secondary-before-fastball rate as a separate covariate, and a per-pitch deviation from a pitcher’s own typical sequencing habits can carry a different sign than the raw, unconditional comparison. This is a real and informative distinction, not a contradiction, but it means the paper’s headline claim describes the unconditional relationship in Table 1, not the fully-adjusted one in Table 6, and that scope should be stated plainly rather than implied.

The modest marginal R^2 values in M2 and M3 are expected: M2 models the swing *residual*, from which location, shape, and count variance have already been removed, so explaining any portion of what remains with arsenal features is meaningful. The conditional R^2 of 0.011 in M3, up from a marginal R^2 of 0.001, shows individual pitcher and batter talent explains more of the remaining variance than arsenal features alone, though the gap is small and shouldn’t be read

Table 7: Mixed effects model variance explained, 2026 NCAA D1. Marginal R2 = fixed effects only; Conditional R2 = fixed + random effects.

Model	R2 Marginal	R2 Conditional
M1: FB Swing Rate	0.240	0.293
M2: FB Swing Residual	0.021	0.036
M3: FB Run Value	0.001	0.011

as strong validation on its own.

10.4 Count Run Values

Table 8: MLB count-based run expectancy. Source: Tango, Lichtman and Dolphin; calibrated to Statcast 2015-2022. Positive values favor the batter.

Count	Run Expectancy
0-0	0.000
0-1	-0.068
0-2	-0.146
1-0	0.033
1-1	-0.035
1-2	-0.113
2-0	0.094
2-1	0.017
2-2	-0.053
3-0	0.179
3-1	0.100
3-2	0.036

11 Conclusion

Traditional pitch metrics are built on a pitch-by-pitch accounting of baseball value. They are precise, reproducible, and widely used, and they miss something systematic.

When a pitcher develops an elite secondary pitch, its value extends well past the moment it's thrown, into every fastball that follows. The hitter who has just processed a 79 mph changeup with 16 inches of fade cannot simply reset when the 94 mph fastball arrives. The timing model is disrupted. The commitment is slightly off. The swing comes a fraction early, or the take comes a fraction late.

I measured that disruption directly: as a behavioral change in swing decisions (the Cognitive Tax), a change in fastball outcomes (the Performance Tax), and a quantity of indirect run value (the Anticipation Tax). The effect holds up across alternative sample definitions and multiple modeling approaches; the permutation test on the raw swing-residual delta specifically did not reach significance in this run (Section 2.2, Appendix A), which tempers how strongly the causal claim below should be read until that discrepancy is investigated further. It is driven more by pitch-to-pitch shock, how much a given pitch differs from the one immediately before it, than by a pitcher's overall, season-long arsenal separation. It is largest in full counts, it compounds when secondary pitches are stacked, and it concentrates in hitters already predisposed to attack.

A secondary pitch's value extends beyond its own CSW rate: it's also expressed in the run value of every fastball thrown to a hitter who knows that pitch exists. A pitcher with an elite changeup is harder to face for two compounding reasons: the changeup itself is hard to hit, and the fastball that follows becomes harder to decide on.

The Anticipation Tax is the run value of that decision difficulty. It has always existed. I've just measured it.

12 Appendix

12.1 A. Permutation Test Detail

Table 9: Permutation test results. The true effect does not fall in the extreme tail of the null distribution in this run – this test alone does not distinguish the observed sequence effect from a random relabeling of the same data.

Statistic	Value
True sequence effect (swing residual delta)	0.041
Placebo mean effect	0.005
Placebo SD	0.077
Permutation p-value	0.595

12.2 B. Model Diagnostics

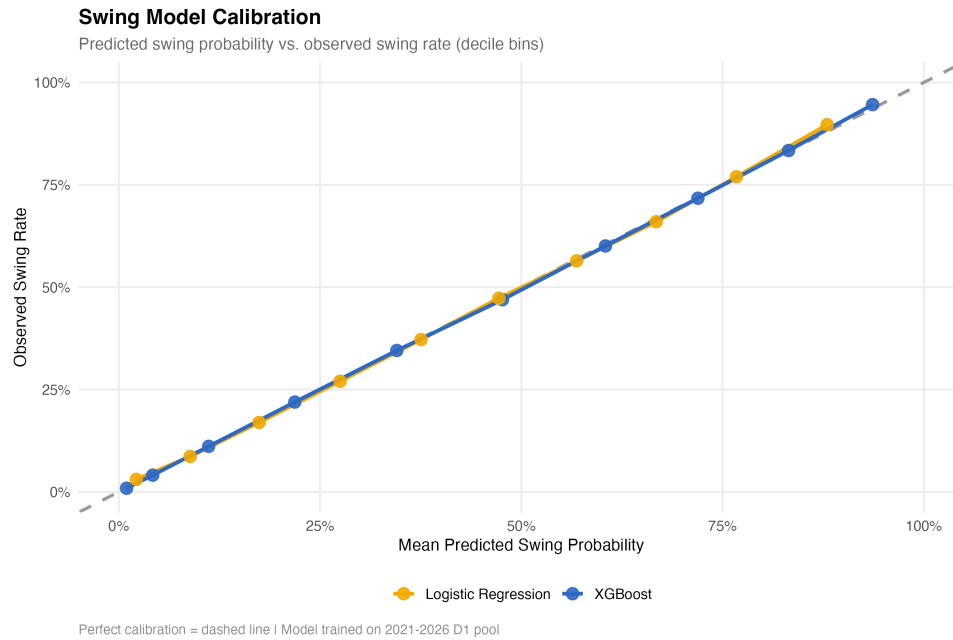


Figure 14: XGBoost swing model calibration on the held-out test set. Predicted probabilities closely track observed swing rates across all deciles, confirming that the swing residual does not systematically overstate or understate disruption in any region of the prediction space.

12.3 C. SHAP Feature Importance

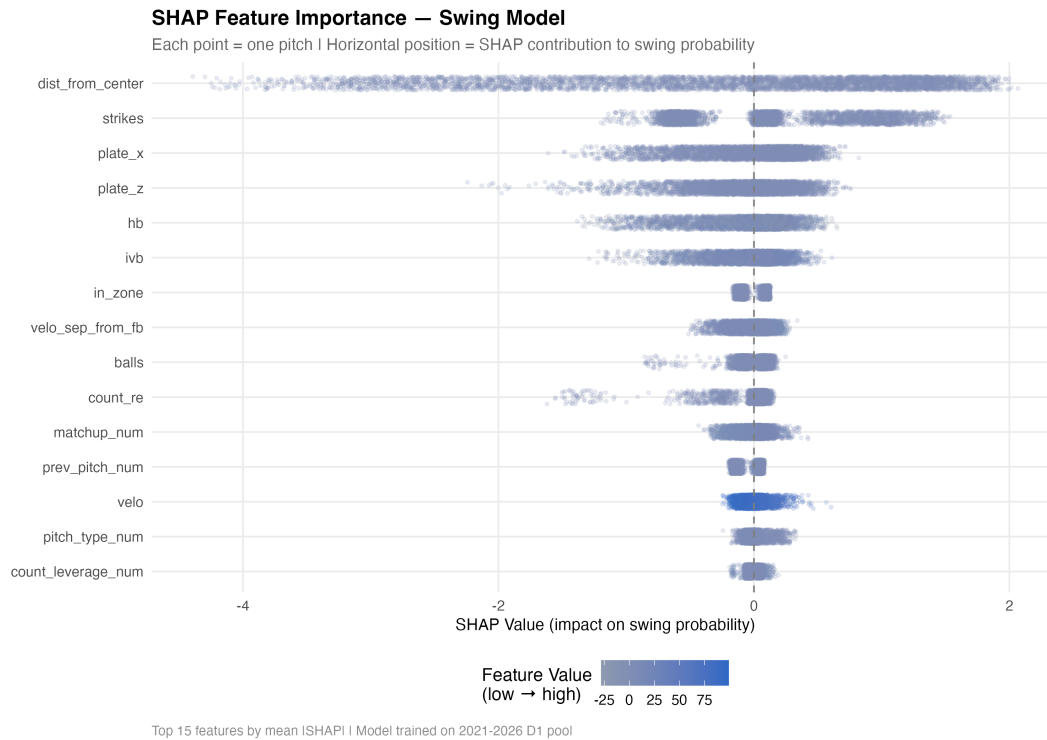


Figure 15: SHAP summary for the XGBoost swing model. Location features dominate as expected. Rolling secondary rate and prev_was_secondary contribute above several pitch shape features, confirming that arsenal context is detectable at the individual pitch level even after location and count are controlled.

12.4 D. Pitch Movement Reference — Right-Handed Pitchers

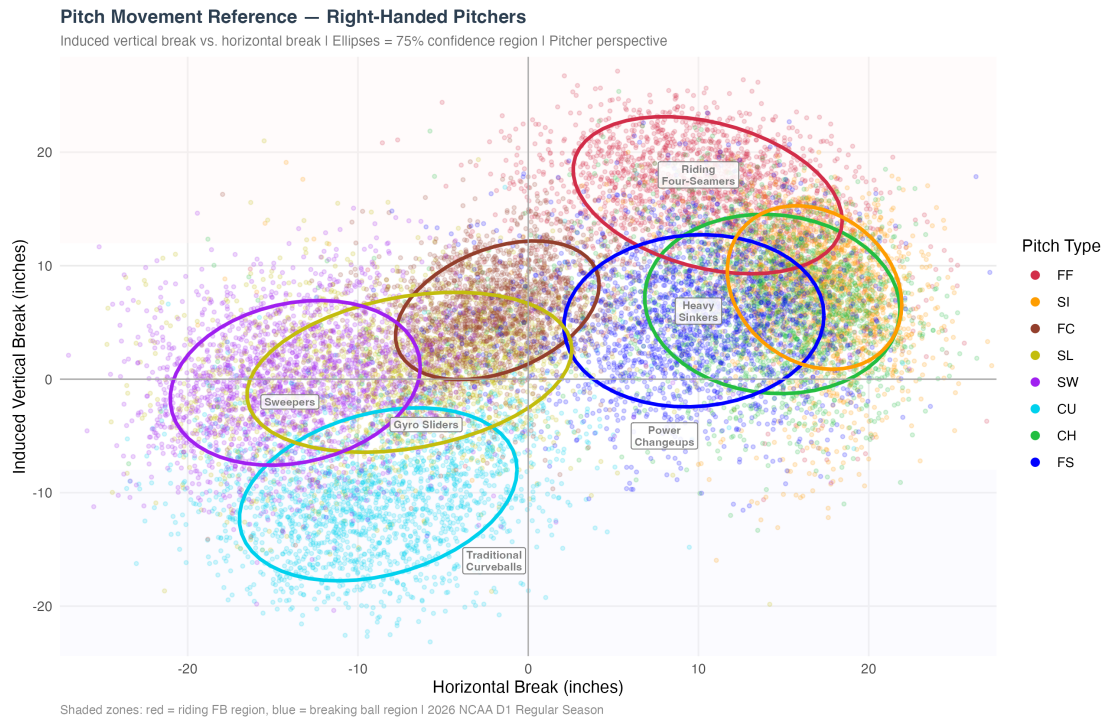


Figure 16: Movement profiles for right-handed pitchers with annotated pitch type regions. Use as a reference when interpreting ACI and mechanism results throughout the report. Shaded zones indicate the riding four-seamer region (top) and breaking ball region (bottom).

12.5 E. Pitch Movement Reference — Left-Handed Pitchers

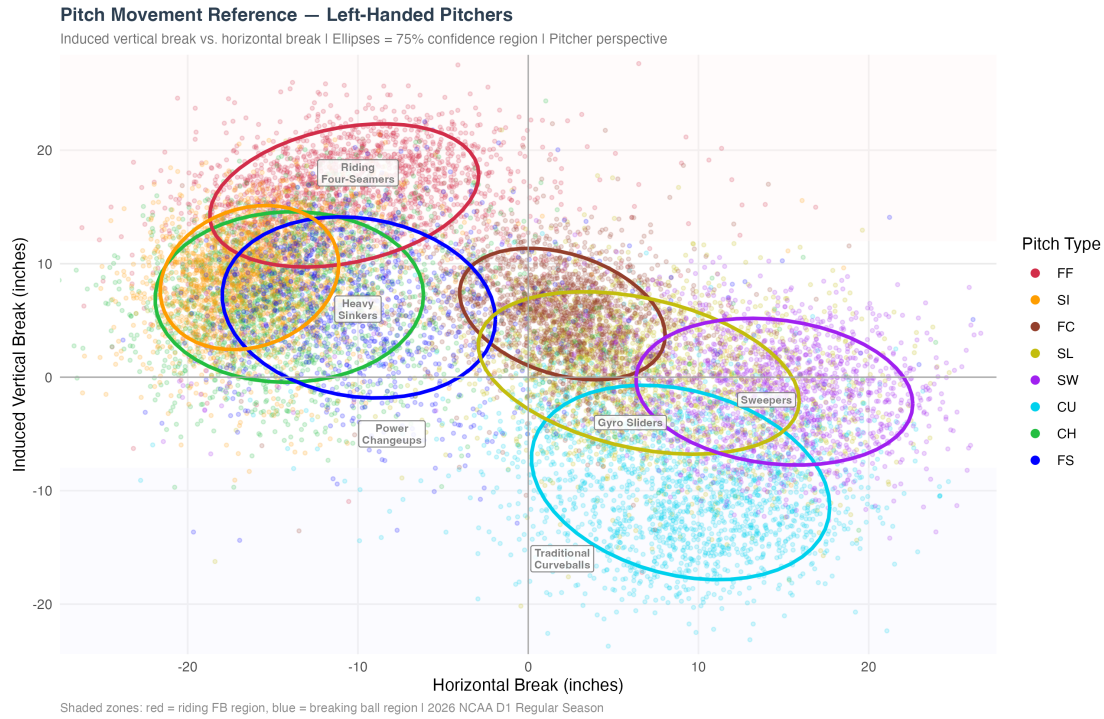


Figure 17: Movement profiles for left-handed pitchers with annotated pitch type regions.

12.6 F. Average Pitch Shape Reference — RHP

Table 10: Average pitch shape by type — Right-Handed Pitchers, 2026 NCAA D1 regular season. Min. 50 pitches per type.

Pitch Type	N	Velo (mph)	IVB (in)	HB (in)	Spin (RPM)	Ext (ft)	CSW%	Whiff%
Curveball	101146	77.8	-9.6	-8.6	2423	5.5	10.9	31.5
Slider	327727	80.5	0.5	-7.2	2388	5.5	13.1	31.7
Sweeper	8551	80.1	-0.3	-13.3	2502	5.5	14.3	35.1
Cutter	97750	84.7	6.0	-1.6	2337	5.8	12.4	26.8
Four-Seam FB	679328	90.0	15.7	10.3	2218	6.0	7.6	17.3
Sinker	90300	89.7	7.7	16.3	2160	5.9	6.9	15.6
Changeup	147493	81.8	6.2	13.3	1704	5.9	15.3	32.9
Knuckleball	385	76.0	5.4	4.8	1873	5.9	6.0	15.9
Splitter	10083	81.4	5.1	9.4	1341	5.7	18.6	39.5

Note: CSW = called strike + whiff rate. Whiff = whiff per swing. IVB = induced vertical break. HB = horizontal break.

12.7 G. Average Pitch Shape Reference — LHP

Table 11: Average pitch shape by type — Left-Handed Pitchers, 2026 NCAA D1 regular season. Min. 50 pitches per type.

Pitch Type	N	Velo (mph)	IVB (in)	HB (in)	Spin (RPM)	Ext (ft)	CSW%	Whiff%
Changeup	63538	80.3	6.9	-13.0	1693	5.8	15.5	33.3
Curveball	33192	76.3	-8.9	8.6	2332	5.4	11.0	32.1
Cutter	22095	83.3	5.4	1.9	2311	5.7	13.7	28.7
Four-Seam FB	233291	88.5	15.7	-10.5	2160	5.9	7.8	18.2
Sinker	33382	89.1	8.7	-15.7	2144	5.9	7.3	17.2
Slider	101589	78.8	0.3	6.6	2323	5.5	13.4	32.5
Splitter	967	80.0	5.9	-9.2	1308	5.5	20.2	42.7
Sweeper	1489	78.4	-1.1	13.5	2477	5.5	12.4	34.4

Note: Same abbreviations as Appendix F.

12.8 H. ACI Leaderboard

Table 12: Top 15 pitchers by Arsenal Contrast Index (ACI), 2026 NCAA D1. ACI = weighted composite of velocity, IVB, and horizontal break separation between fastball and best secondary pitch.

Pitcher	Throws	Pitches	Types	Entropy	VSep (mph)	Sec%	Best Sec	ACI
King, Ryan	R	933	3	0.65	19.6	24%	SL	3.19
Mendez, Connor	R	355	4	1.17	12.3	44%	CH	2.86
Sanford, Will	R	1487	4	0.96	14.4	36%	CH	2.61
Schulz, Jakob	L	327	4	1.28	11.9	56%	FS	2.50
Oakley, Reese	R	351	4	1.22	11.0	49%	CH	2.49
Shadek, Justin	R	534	4	0.83	15.4	28%	CU	2.46
Jensen, Sawyer	R	1243	4	1.10	15.3	44%	CH	2.41
Poncini, Michael	R	315	3	0.96	14.7	42%	CU	2.39
Nichols, Gunnar	R	762	3	0.96	13.3	41%	CH	2.36
Flukey, Cameron	R	407	3	0.99	16.0	45%	CU	2.35
McNaught, Ray	R	210	2	0.66	16.9	37%	CU	2.30
Ludeman, Matt	R	201	2	0.58	17.8	27%	CU	2.29
Foxson, Tate	R	303	3	0.71	17.0	33%	CU	2.27
Dougan, Justin	L	137	4	0.80	15.9	26%	CU	2.25
Stephens, Brandon	R	164	2	0.49	15.8	20%	CU	2.24

ACI = Arsenal Contrast Index. VSep = velocity separation between FB and best secondary (mph). Entropy = Shannon entropy of pitch mix. Types = distinct pitch types thrown.

12.9 I. Data Provenance & Scope Correction

This report’s data pipeline was revised after an earlier version of this project was found to have an unintended scope issue, described here for transparency.

What happened. The original analysis pulled TrackMan data from a single flat folder assumed to contain only NCAA Division I games for the 2026 season. It didn’t: the folder mixed several levels of NCAA competition (D1, D2, D3, JUCO, and others) together under what was believed to be a clean D1-only sample. Every script ran correctly against the data it was given, so this wasn’t a coding bug in the usual sense. It was a scope error. The data included levels of competition the analysis was never intended to describe, which would have understated the quality of

competition represented in every leaderboard, arsenal profile, and mixed-effects estimate in the report.

How it was found. The TrackMan archive was later reorganized on external storage into an explicit `NCAA Data/{YEAR}/{LEVEL}/` folder structure. That made it clear Division I data was available as its own distinct subfolder, separable from other levels, across six seasons (2021–2026) rather than one.

The correction. The pipeline was rewritten around two deliberately different scopes:

1. **Training scope:** `01_data_cleaning.R` now loads *only* the D1 subfolder, explicitly, for each year 2021-2026, and tags every row with the season and level it was read from at load time, before any data from different years or levels is combined. If a year is missing a D1 folder, the pipeline raises a warning rather than silently substituting another level. This pooled, D1-only, multi-season dataset fits the swing probability model (Section: Swing Decision Model), since a larger and more varied training pool produces a more generalizable model than a single season would.
2. **Reporting scope:** immediately after the swing model is fit and scored (`04_swing_model_xgboost.R`), the pipeline filters down to `season == 2026` before any further script runs. Every subsequent script, arsenal feature construction, mixed effects models, the Anticipation Tax leaderboard, mechanism analysis, hitter segmentation, and all figures and tables, operates exclusively on this 2026 D1-only subset.

Why this split matters. A pitcher-level classification like “elite secondary pitch” is only meaningful relative to a specific, well-defined population. Computing it against a population that unintentionally blended multiple levels of competition would misclassify pitchers relative to the D1 field they actually face, and blending multiple seasons together would describe a pitcher’s career rather than their 2026 season specifically. The corrected pipeline recomputes every reporting-scope classification, pitch mix entropy, velocity separation, elite-secondary status, best secondary pitch, fresh, from 2026 D1 pitches only, rather than reusing any statistic computed on the broader training pool. The one exception is the swing probability model itself, intentionally trained on the wider 2021-2026 D1 pool and then *applied* to 2026 pitches, the same way a model trained on more historical data would be applied to score a current season in any standard machine learning workflow.

What this means for interpreting this report. All headline figures here were recomputed under the corrected scope and may differ, in some cases substantially, from any earlier version of this analysis that used the mixed-level sample. That recomputation strengthens the analysis rather than weakening it: the underlying behavioral mechanism (fastball value increasing after a secondary pitch) has nothing to do with which levels of competition happened to be present in an earlier data pull, and re-establishing it on a clean, explicitly-scoped 2026 D1 sample is simply the more defensible test of the claim.